Name: _____ Class and student number: _____

Give out this part only after the original investigation answers have been submitted.

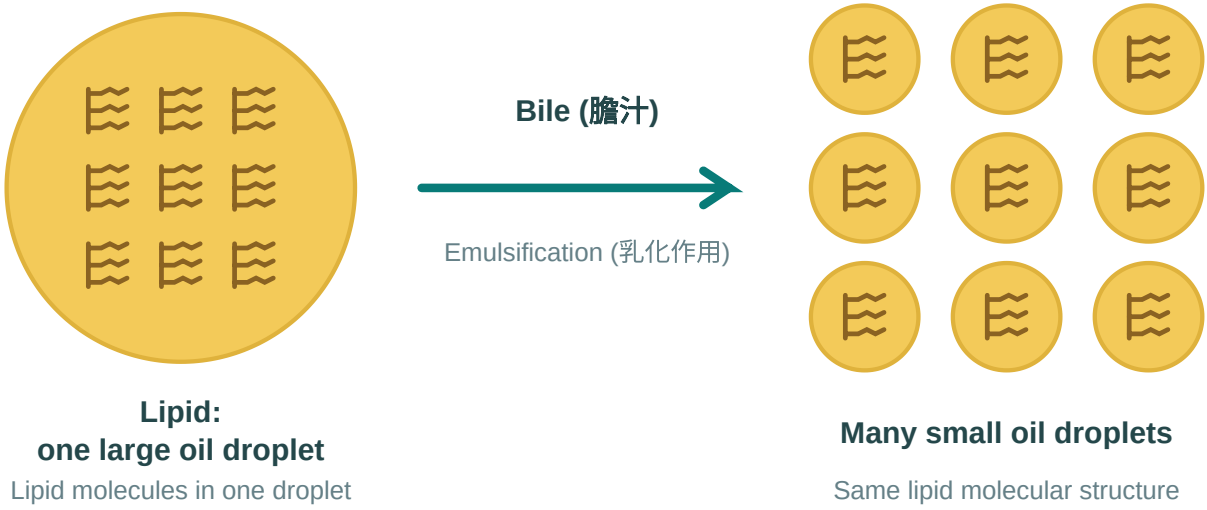
Key learning points

You have now explored the functions of digestive juices X and Y.

1. Digestive juice X is bile (膽汁). It is produced in the liver and stored in the gall bladder (膽囊). Bile (膽汁) contains no digestive enzymes. Through physical digestion (物理消化), it emulsifies lipids (乳化脂質) into small oil droplets, increasing their surface area so chemical digestion (化學消化) can work more efficiently.

2. Digestive juice Y is lipase (脂肪酶). It catalyses the chemical digestion (化學消化) of lipids, breaking them down into smaller molecules of glycerol (甘油) and fatty acids (脂肪酸) to aid absorption. Lipase (脂肪酶) is a protein; high temperature causes denaturation, so it loses its function.

01 • Physical digestion (物理消化): emulsification (乳化作用), with no change in molecular structure



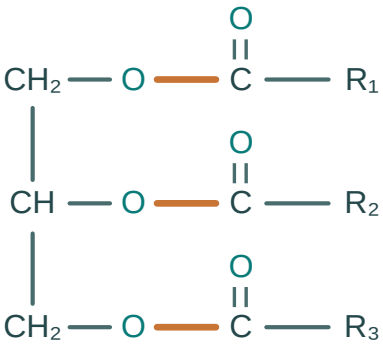
Same lipid amount; greater total surface area

Only the size of the oil droplets and how spread out they are change; lipid molecules are not broken down. Small droplets increase the total surface area. The diagram does not show the real sizes.

Give out this part only after the original investigation answers have been submitted.

02 • Chemical digestion (化學消化): hydrolysis forms different molecules

Lipid: triglyceride (甘油三酯)



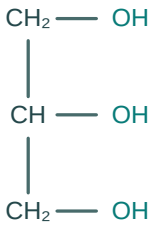
Orange bonds: where hydrolysis happens

+ $3H_2O$ (water)

lipase (脂肪酶)

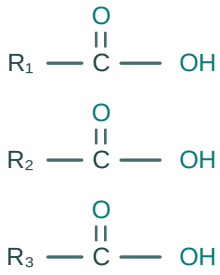
Hydrolysis

glycerol (甘油)



+

3 fatty acids (脂肪酸)



Triglyceride (甘油三酯) + $3H_2O$ ⟶ Glycerol (甘油) + 3 fatty acids (脂肪酸)

R_1, R_2 and R_3 show the hydrocarbon chains of fatty acids (脂肪酸). Hydrolysis changes chemical bonds to form glycerol (甘油) and fatty acids (脂肪酸). Lipase (脂肪酶) is a catalyst and is not used up in the reaction. This diagram uses the simple complete-hydrolysis model commonly taught in this course.

What the evidence can tell us

What the evidence can tell us: this simulation uses set appearances to show emulsification (乳化作用) and enzyme action. A clear mixture does not directly show that glycerol (甘油) or fatty acids (脂肪酸) have formed, or measure digestion speed. Real experiments need other chemical tests and repeated tests. In the heating comparison, we assume that boiled liquids are cooled to the same reaction temperature before use.

Learning reflection

Do your observations support your first idea to test? Name the groups you compared and what you saw. Use the key learning points to improve your explanation.

Refer to your original idea to test and reason saved in Part A. You do not need to rewrite them from memory.